

electrodes that are currently used in research and clinical applications, the nanotubes can create shortcuts between the two extremes of a neuron, resulting in enhanced neuronal excitability.

For Professor Henry Markram, project leader, the most exciting part of his research is putting together the hundreds of thousands of small pieces of data that his lab has collected over the past 15 years, and seeing what a microcircuit of the brain looks like.

“When we first switched it on, it already started to display some interesting emergent properties,” he says. “But this is just the beginning because we know now that it is possible to build it. As we progress we are learning about design secrets of our brains which were unimaginable before. In fact the brain uses some simple rules to solve highly complex problems and extracting each of these rules one by one is very exciting. For example we have been surprised at finding simple design principles that allow billions of neurons to connect to each other. I think we will understand how the brain is designed and works before we have finished building it.”

Meanwhile, a different group of scientists at the California Institute of Technology has successfully made an artificial neural network out of DNA, creating a circuit of interacting molecules that can recall memories based on incomplete patterns, just as a human brain can.

While this proof-of-principle experiment shows the promise of creating DNA-based networks that can—in essence—think, this neural network is limited, the researchers say. The human brain consists of 100 billion neurons, but creating a network with just 40 of these DNA-based neurons – 10 times larger than the demonstrated network – would be a challenge, according to the researchers.

Markram, however, is optimistic and believes that the human brain could be replicated in the next 10 years: “I absolutely believe it is technically and biologically possible.”

“The brain is, of course, extremely complex because it has trillions of synapses, billions of neurons, millions of proteins, and thousands of genes. But they are still finite in number. Today’s technology is already highly sophisticated and it allows us to reverse engineer the brain rapidly.”

Love and sex

According to David Levy – yes, the same chess grandmaster who had once predicted that no computer would beat him – we’re fast approaching an age where it would be possible for humans to fall in love with robots